

Adhesive Forces in Droplet Kinetic Friction on Liquidlike Surfaces

Glen McHale^{1,*}, Sara Janahi¹, Hernán Barrio-Zhang¹, Yaofeng Wang¹, Jinju Chen²,
Gary G. Wells¹ and Rodrigo Ledesma-Aguilar¹

¹*Institute for Multiscale Thermofluids, School of Engineering, The University of Edinburgh,
Edinburgh, EH9 3FB, United Kingdom*

²*Department of Materials, Loughborough University, Loughborough, LE11 3TU, United Kingdom*



(Received 21 June 2025; revised 23 August 2025; accepted 16 October 2025; published 12 November 2025)

Kinetic frictional forces resisting droplet motion often appear to be separate from surface wettability and adhesive forces. Here we show that such friction arises from a simple combination of the contact angle hysteresis and adhesive force. We show theoretically, and confirm using tilt angle experiments of droplets on liquidlike surfaces, the dependence of the coefficient of droplet-on-solid kinetic friction on system parameters. We also show that a molecular kinetic-type model can describe the observed nonlinear velocity-force relationship. Our findings provide a fundamental understanding of the relationship between droplet-on-solid friction, and wettability and liquid adhesion.

DOI: [10.1103/ljbc-k193](https://doi.org/10.1103/ljbc-k193)

Introduction—From droplets of rain on a window to droplets of coffee on a table, small sessile droplets resting on surfaces are ever present in society. Understanding these situations is underpinned by the studies of Thomas Young in 1805 which related the three interfacial tensions separating the solid, liquid, and gas interfaces from the observable angle of contact at the three-phase contact line where the droplet meets the supporting solid [1–3]. However, if Young’s Law held perfectly for practical surfaces, a small droplet of rain would always slide down a window, and drying droplets would never leave ring stains [4]. Describing such problems arising from contact line pinning leads to the apparent abandonment of Young’s theoretical equilibrium contact angle and the use of empirically observed advancing and receding contact angles [5]. These two angles describe the possible range of contact angles—the contact angle hysteresis—between which the measured static contact angle will be found in any deposition process. Contact angle hysteresis is a characteristic of any practical surface and represents the variation from a perfectly smooth and homogeneous surface due to roughness and surface chemistry. The force pinning a static droplet can be considered a static droplet-on-solid friction force with the resistance to maintaining droplet motion being the corresponding dynamic droplet-on-solid friction in analogy to the separation of static and kinetic regimes for one solid sliding on another [6–8].

Recently, droplet-on-solid kinetic friction has been studied through measuring forces on cantilevers as a substrate is displaced [9–11]. It has been suggested that an empirical law relating the droplet-on-solid friction to the speed of droplet motion might apply, and a dimensionless friction coefficient, β , has been suggested as a material parameter [11]. Whilst practical surfaces always exhibit some heterogeneity, there has been significant progress in techniques to create smooth ultralow contact angle hysteresis surfaces using covalently attached liquidlike polymer chains [12–14]. It has also been shown that droplets of water on polydimethylsiloxane (PDMS) liquidlike surfaces can have similar static contact angles, but significantly different speeds of motion once static friction is overcome [15,16]. This aspect is a new paradigm for droplets and is challenging for established concepts. Such surfaces where kinetic friction can be converted from high to low, offer a unique opportunity to investigate the relationship between wettability, and droplet-on-solid kinetic friction and droplet adhesion.

Here, we provide a first-principles approach to defining coefficients of droplet-on-solid friction from the ratio of frictional and adhesive forces with excellent agreement with experimental observations on low and high droplet-on-solid kinetic friction surfaces. We also show the experimentally observed droplet kinetic friction-speed relationship for droplets of water on liquidlike surfaces is nonlinear and can be described by a molecular kinetic theory (MKT) type model [17,18], which inherently encapsulates the droplet adhesion. The first order expansion of our MKT type model provides a linear approximation which incorporates the previous empirical law.

Droplet-on-solid friction and the adhesive force—For a solid object sliding on another solid the weight of the object

*Contact author: glen.mchale@ed.ac.uk

Published by the American Physical Society under the terms of the [Creative Commons Attribution 4.0 International license](https://creativecommons.org/licenses/by/4.0/). Further distribution of this work must maintain attribution to the author(s) and the published article’s title, journal citation, and DOI.

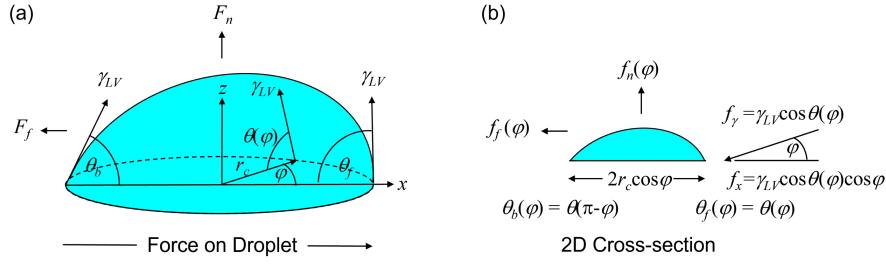


FIG. 1. Response of a droplet to an applied force. (a) Normal component of liquid-vapor interfacial tension force, F_n , and the in-plane frictional force, F_f , acting on a droplet distorted in the x - z plane by an applied force. (b) Interpretation of forces on the three-phase contact line as arising from two-dimensional cross sectional profiles each with contact length $2r_c \cos \varphi$ and front and back contact angles $\theta(\varphi)$ and $\theta(\pi - \varphi)$, respectively, at positions parametrized by the angle φ along the contact line.

is translated into frictional forces opposing motion through static and kinetic coefficients of friction following Amontons' law [19,20]. However, for a droplet whose size is smaller than the capillary length $l_c = (\gamma_{LV}/\rho g)^{1/2}$, where γ_{LV} and ρ are the liquid-vapor interfacial tension and the density of the liquid, respectively and g is the acceleration due to gravity, capillary forces dominate over the force due to gravity. We therefore define the coefficient of droplet friction, μ , as the ratio of the frictional force, F_f , to the droplet adhesive force from the normal component, F_n , of the capillary force using an Amontons-like law [21],

$$\mu \equiv \frac{F_f}{F_n}, \quad (1)$$

where μ is either a static coefficient, μ_s , or a kinetic coefficient, μ_k , respectively, when the droplet is static or in motion. We now assume a small distortion from a spherical cap cross sectional profile shape in the x - z plane [Fig. 1(a)] due to an applied force in the x direction, and use a smooth and continuous parametrization of the contact angle, $\theta(\varphi)$ where φ ranges from 0 to $\pm\pi$, around a circular three-phase contact line from its maximum value at the front, $\theta(0) = \theta_f$, to its minimum value at the back, $\theta(\pi) = \theta_b$,

$$\cos \theta(\varphi) = \cos \theta_f + \frac{1 - \cos \varphi}{2} (\cos \theta_b - \cos \theta_f). \quad (2)$$

Equation (2) is the simplest continuous parametrization and includes the first nonconstant term in a Fourier cosine series description of the surface tension force dependence on the contact angle. The in-plane component of the solid-liquid interfacial tension force per unit length directed toward the center of the droplet at a point on the contact line is $f_\gamma = \gamma_{LV} \cos \theta(\varphi)$, and the x component is $f_x = \gamma_{LV} \cos \theta(\varphi) \cos \varphi$. Integrating around the contact line gives a Furmidge-type equation for the frictional force [22–24], as shown in the Supplemental Material, Sec. S1 [25]:

$$F_f = 2kr_c \gamma_{LV} (\cos \theta_b - \cos \theta_f), \quad (3)$$

where the shape factor is $k = \pi/4 \approx 0.785$ for a circular contact area. For a droplet on the cusp of motion, the frictional force is the contact pinning force, and the front and back contact angles in Eq. (3) are equal to the static advancing and receding contact angles, $\theta_f = \theta_a$ and $\theta_b = \theta_r$, respectively. For a droplet in motion, the frictional force is given by the dynamic front and back contact angles, which are dependent on the speed of droplet motion. A similar evaluation of the normal forces around the circular three-phase contact line gives $F_n = 2\pi r_c \gamma_{LV} \sin \theta_{ave}$, where $\theta_{ave} = (\theta_f + \theta_b)/2$ (Supplemental Material, Sec. S1 [25]).

Physically, the shape factor, k , can be interpreted as a weighting factor for combining the net force in the x - y plane in the direction of distortion arising from cross sectional profiles taken from one side of the droplet to the other, each with contact angles $\theta_f = \theta(\varphi)$ and $\theta_b = \theta(\pi - \varphi)$ [Fig. 1(b)]. Importantly, it does not represent an elongation or distortion of the contact area, which is often its interpretation in the literature on the Furmidge equation. However, if there is a known distortion of the three-phase contact line from circular, a shape factor could still be calculated; an example for an elliptical contact line is given in the Supplemental Material, Sec. S1 [25]. The methodology for assessing a coefficient of droplet friction using Eqs. (1) and (2) could also be applied to a wider class of contact lineshapes, such as those reported by Le Grand *et al.* [26].

From our results, the coefficient of droplet-on-solid friction only depends on the two contact angles (θ_f, θ_b) and the shape factor, k . Alternatively, we can use the average contact angle θ_{ave} , and the contact angle difference $\Delta\theta = (\theta_f - \theta_b)$, to identify the difference of cosines in Eq. (3) and expand the sines in the ratio of forces to first order, so it becomes

$$\mu = \frac{k\Delta\theta}{\pi}. \quad (4)$$

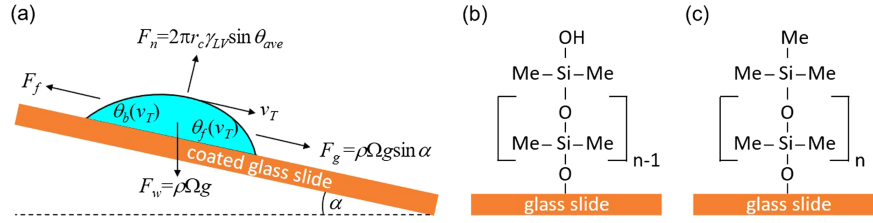


FIG. 2. Measurements of droplet kinetic friction on hydrophobic PDMS-based liquidlike surfaces. (a) Tilt angle experiments with friction and normal adhesive forces measured under constant speed motion. (b) Ultralow hysteresis covalently attached OH-terminated PDMS chains providing a coating with high droplet kinetic friction. (c) Methyl-terminated PDMS chains obtained by using a molecular capping procedure to convert the surface in (b) into a low droplet kinetic friction surface.

In the context of Eq. (1), wettability defined through the average contact angle also defines the normal component of droplet capillary force and hence the adhesion of the droplet to the surface. The coefficient of friction, Eq. (4), then translates the liquid-solid adhesive force normal to the surface into friction in the plane of the surface. The preceding equations describing droplets can be applied whether a droplet is static or in motion and so used to define both static and kinetic coefficients of friction, μ_s and μ_k . The difference is whether the average contact angle is a static contact angle θ_s , approximating the one expected from Young's law, i.e., $\cos \theta_e = (\gamma_{SV} - \gamma_{SL})/\gamma_{LV}$ [1,27], and the advancing θ_a and receding θ_r contact angles are assumed, or whether the average contact angle is a dynamic contact angle θ_d and the dynamic front θ_f and back θ_b contact angles are used. We would not expect the coefficient of droplet-on-solid kinetic friction defined through Eq. (1) to be a material parameter because the difference $\Delta\theta$ depends on the speed of droplet motion. It can, however, be considered a system parameter dependent on the interactions between the liquid and the surface. In this context, changes in interaction due to, e.g., slide electrification [8], can be expected to alter the kinetic friction, but we would expect our unified framework for the relationship between droplet friction, wettability, and liquid adhesion to remain valid.

From the perspective of surface design, a low droplet-on-solid static friction surface can be achieved by reducing the normal component of the capillary force or by reducing the contact angle hysteresis. Thus, both low and high droplet adhesion surfaces can be created with low droplet-on-solid static friction. This can be shown by using the spherical cap volume, Ω , to relate the contact radius to the contact angle and replacing the contact angle by the work of adhesion, $W_{adh} = \gamma_{LV}(1 + \cos \theta_e)$ in the normal component of the capillary force (Supplemental Material, Sec. S2 [25]),

$$F_n = 2\pi\gamma_{LV} \left(\frac{3\Omega}{\pi \left(1 + \frac{W_{adh}}{\gamma_{LV}} \right)} \right)^{1/3} \left(\frac{W_{adh}}{\gamma_{LV}} \right) \left(2 - \frac{W_{adh}}{\gamma_{LV}} \right)^{1/3}. \quad (5)$$

In the limit of a perfectly superhydrophobic ($\theta_e \rightarrow 180^\circ$) or a perfectly hydrophilic surface ($\theta_e \rightarrow 0^\circ$), the work of adhesion and the normal component of the capillary force both vanish. Hence, in these limiting cases, including the perfectly hydrophilic one corresponding to a film-forming surface, the droplet-on-solid friction is zero irrespective of the contact angle hysteresis, $\Delta\theta_{CAH} = \theta_a - \theta_r$. The maximum in the normal adhesive force for a spherical cap droplet of fixed volume occurs for a hydrophilic surface with a contact angle $\theta_e \approx 65.5^\circ$ corresponding to a work of adhesion $W_{adh} = 2^{1/2}\gamma_{LV}$ (Supplemental Material, Fig. S2 [25]). The maximum occurs at a contact angle below 90° because of the dependence of the contact radius, r_c , on the contact angle for a fixed volume. For a surface on which there is low droplet pinning, minimizing the contact angle hysteresis is critical to minimize the coupling of the adhesive force into a frictional force.

Experiments on liquidlike surfaces—To explore whether the coefficient of kinetic friction defined by Eq. (1) is proportional to the difference between the dynamic front and back contact angles [Eq. (4)], we created hydrophobic liquidlike surfaces on glass substrates and conducted tilt angle experiments [Fig. 2(a)] (Supplemental Material, Sec. S3 [25]). A “grafting-from” acid-catalyzed polycondensation procedure of dimethyldimethoxysilane was used to covalently attach hydroxy-terminated PDMS chains [Fig. 2(b)], i.e., a slippery omniphobic covalently attached liquidlike surface (SOCAL), with ultralow contact angle hysteresis $\Delta\theta_{CAH} \sim 2^\circ$ [28,29]. Although these surfaces have ultralow contact angle hysteresis (static friction), once droplets are in motion they move slowly (high kinetic friction) [15]. This has previously been attributed to the interaction between water and the silanol groups in the hydroxy-terminated PDMS chains [16]. We therefore used a molecular-capping (methylation) procedure, which is believed to convert the silanol-termination of the PDMS chains to trimethylsilyl-termination (c-SOCAL), to create a set of surfaces with lower kinetic friction [Fig. 2(c)] [16,30]. The effectiveness of this vapor-phase capping procedure depends on the process parameters, particularly the relative humidity (RH) because of the reactivity of the chlorotrimethylsilane used in the methylation procedure to water vapor. We created surfaces using RH = 30%, and

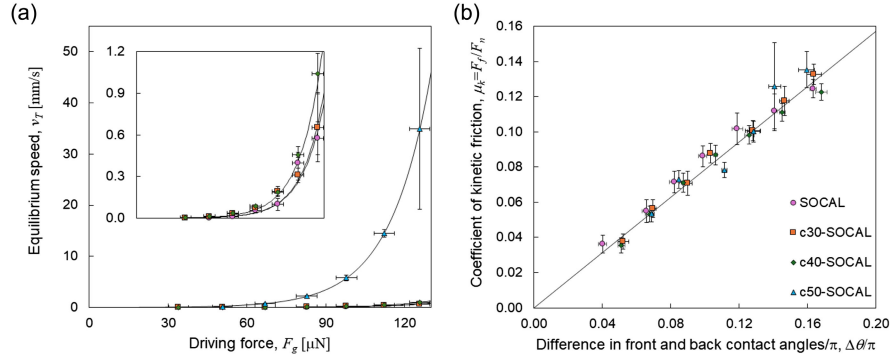


FIG. 3. (a) Equilibrium speeds of motion of 20 μl droplets on high (SOCAL, c30-SOCAL, and c40-SOCAL) and low (c50-SOCAL) droplet kinetic friction liquidlike surfaces at substrate tilt angles up to 40° to the horizontal; solid lines are fits of the speed to a molecular-kinetic theory type model [Eq. (8)], and the inset shows details for the higher friction surfaces using a smaller range for the y axis and the same range for the x axis. (b) Linearity of the coefficient of droplet-on-solid kinetic friction, μ_k , determined using measurements of r_c , θ_f , and θ_b ; solid line is the predicted slope of $k = \pi/4$.

40% and 50% (c30-SOCAL, c40-SOCAL, and c50-SOCAL) and found a significant reduction of the kinetic friction for the c50-SOCAL surface. The capping process makes little difference to the measured hydrophobicity of the samples (the static contact angle increases from $\sim 103^\circ$ to $\sim 106^\circ$), although the contact angle hysteresis increases from $\sim 2^\circ$ to $\sim 5^\circ$ on the c50-SOCAL surface, which we interpret as being due to the HCl by-product of the capping process on the surface homogeneity.

We recorded the motion of water droplets on surfaces tilted at a range of angles, α , up to 40° and measured the front and back dynamic contact angles, and the steady state speed v_T , at which the frictional force, F_f , balances the component of gravitational force along the surface, $F_g = \rho\Omega g \sin \alpha$. Figure 3(a) shows data for the observed speed of 20 μl droplets for frictional forces, $F_f = F_g$, up to around 126 μN ; the kinetic frictional force is a nonlinear function of the speed. For droplets on the high (SOCAL) and low (c50-SOCAL) kinetic friction surfaces tilted at 40° , the speeds are $v_T = 0.58 \text{ mm s}^{-1}$ to $v_T = 35 \text{ mm s}^{-1}$, respectively. This is a remarkable increase induced by the capping procedure despite the increase in the static contact angle hysteresis (droplet-on-solid static friction). We did not observe any significant deviation from a circular contact area shape (Supplemental Material, Sec. S3 [25]). To explore whether Eq. (4) is valid experimentally, we calculated the normal component of the surface tension force, F_n , from the average of the front and back contact angles and the observed contact radius of the droplet. In Fig. 3(b) we plot the coefficient of droplet-on-solid kinetic friction, μ_k , determined by the ratio of forces in Eq. (1), and $\Delta\theta/\pi$ using the measured difference in front and back contact angles over a range of tilt angles up to $\alpha = 40^\circ$. The data follow a linear relationship with an experimentally determined slope of $k = 0.79 \pm 0.04$ in excellent agreement with the predicted value of $k = \pi/4 = 0.785$ from Eq. (3). Prior values derived in the literature for the shape

factor k lie between $2/\pi = 0.637$ [31] and $24/\pi^3 = 0.774$ [32]. However, these have assumed either noncontinuous contact angle distributions along the contact line and/or an elongated droplet contact lineshape. In our approach, the shape factor k is a factor allowing conversion from a two-dimensional model profile as shown in Fig. 1(b) to the three-dimensional droplet shape as shown in Fig. 1(a). Any distortion of the contact area shape from circular will result in a different value for k but, providing reflection symmetry is retained in the x - z axis, Eq. (4) will remain valid.

We now consider the relationship between the dynamic front and back contact angles, θ_f and θ_b , and the droplet speed, v_T . Both contact angles change with speed, but the back contact angle is more sensitive to the speed [Fig. 4(a)]. The back contact angle for droplets moving on the low kinetic friction c50-SOCAL surface has a much smaller change than for the other higher kinetic friction surfaces. However, when the data are considered using the coefficient of droplet-on-solid kinetic friction, μ_k , and plotted using a logarithmic axis, the shape of the observed curves on all surfaces is similar [Fig. 4(b)]; normalizing the velocities on each surface by a factor, v_s , allows the data to be overlaid when plotted using $\log_{10}(v_T/v_s)$ [Fig. 4(b), inset]. Thus, in these tilt angle experiments, similar values of coefficients of droplet-on-solid friction are achieved by droplets of the same volume on different liquidlike surfaces tilted at the same angle, but to do so the droplet has different terminal speeds and, hence, different front and back contact angles.

A MKT-type model—It has previously been suggested that MKT may apply to the motion of contact lines on liquidlike surfaces [15,16] rather than the hydrodynamic theory of Cox-Voinov [3,33,34]. To understand the non-linear response of a droplet to the unbalanced capillary force driving the motion and the droplet-on-solid friction, we now develop a modified MKT-type model formulated using an activation energy, $W(\varphi)$, which varies around the

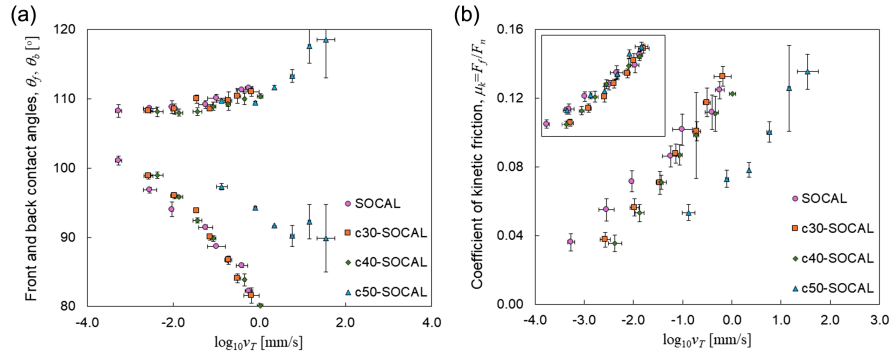


FIG. 4. (a) Velocity dependence of dynamic front and back contact angles for 20 μl droplets on high (SOCAL, c30-SOCAL, and c40-SOCAL) and low (c50-SOCAL) droplet kinetic friction liquidlike surfaces at substrate tilt angles up to 40° to the horizontal. (b) Velocity dependence of coefficient of droplet-on-solid kinetic friction; inset uses a $\log_{10}(v_T/v_s)$ axis, where v_s are the parameters from the MKT model, to overlay the data.

three-phase contact line (Fig. 5). At the contact line the liquid molecules must be in a dynamic equilibrium with both forward and backward displacements (jumps). Following the original MKT model [17,18], the average distance of a molecular displacement of the contact line corresponds to the average distance between adsorption sites on the surface, λ , and the displacement occurs with an average frequency K_o . However, we differ from the original model in defining the activation energy using the dynamic equilibrium contact angle and a local contact angle, parametrized by Eq. (2) to capture the variation around the contact line bounded between the dynamic front and back contact angles,

$$W(\varphi) = \lambda^2 \gamma_{LV} (\cos \theta_d - \cos \theta(\varphi)). \quad (6)$$

The advantage of this definition is that it incorporates the friction-adhesion concept for the motion of the three-phase contact line because for small differences in contact angles, $W(\varphi) \approx \lambda^2 \gamma_{LV} \Delta\theta(\varphi) \sin \theta_d$, where $\Delta\theta(\varphi) = (\theta(\varphi) - \theta_d)$. Using the normal component of the capillary force this gives $w(\varphi) \approx \lambda^2 \mu(\varphi) f_n$, where $\mu(\varphi) = \Delta\theta(\varphi)$ is the local coefficient of droplet-on-solid friction for forward and backward displacements at the contact line position

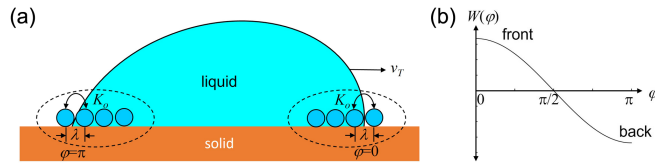


FIG. 5. Schematic showing the molecular-kinetic theory type picture for the displacement of molecules of liquid at the three-phase contact line of the droplet. (a) Motion of molecules at the microscopic level between adsorption sites with average separation of λ at an average frequency of K_o . (b) The energy barrier for forward and backward motion of molecules is proportional to the local coefficient of friction and the normal component of the liquid-vapor interfacial force.

parametrized by φ . The difference between forward and backward jumps around the contact line then gives the net speed of motion of the contact line,

$$v_T = \frac{1}{2\pi r_c} \oint 2\lambda K_o \sinh \left[\frac{W(\varphi)}{k_B T} \right] \cos \varphi d\varphi, \quad (7)$$

which gives (Supplemental Material, Sec. S4 [25])

$$v_T = v_s \sinh \left[\frac{\lambda^2 F_g}{4k_B T} \right] F_g > F_s, \quad (8)$$

where $v_s = \lambda K_o$, k_B is the Boltzmann constant, T is the temperature, and F_s is the static droplet friction force. We fit the data in Fig. 3(a) using a value of $\lambda = 1.33$ nm for all datasets and $v_s = 0.37, 0.32, 0.68$, and 20.5 $\mu\text{m/s}$ for the SOCAL, c30-SOCAL, c40-SOCAL, and c50-SOCAL surfaces, respectively. These fits suggest molecular displacements with average frequencies $K_o = 282, 242, 513$, and 15442 Hz and a distance for displacement, λ , in the nanometer range which is physically plausible. Here, the impact of the molecular capping on droplet-on-solid friction arises through the change in average frequency causing a ca. 55-fold increase in the prefactor v_s . A possible physical interpretation is that the silanol groups on OH-terminated PDMS act as trapping sites, which retain molecules for longer than other groups and so decrease the average molecular jump frequency.

We now consider the consistency between our coefficient of friction and an alternative empirical definition of a friction coefficient, β , given by $F_f = F_o + 2\beta r_c \eta v_T$, where F_o is the extrapolation to zero speed, suggested by Li *et al.* [8,11]. This linear relationship between the friction force and the droplet speed suggested is not an accurate description of our data in Fig. 3(a). However, a first order expansion of Eq. (8) gives a linear relationship, and the models can be matched in this limit by defining

$$\beta = \frac{\pi k_B T}{2\eta K_o \lambda^3}, \quad (9)$$

where we have used $k = \pi/4$ for a droplet with a circular contact area. This suggests $v_s = \pi k_B T / 2\eta \beta \lambda^2$ and so scales with the inverse of viscosity as expected in the original MKT theory by using the Eyring theory for viscosity [35]. This scaling also means a characteristic speed for liquid motion, $v^* = \gamma_{LV}/\eta$, arises naturally in the first order expansion of Eq. (8) from the surface tension dependence of the energy barrier.

Conclusions—In summary, our approach shows droplet-on-solid kinetic friction can be related directly to the droplet adhesive force arising from wettability of a surface. It provides an understanding of the shape factor in the frictional force as a translation from the forces in a two-dimensional cross sectional droplet profile to the three-dimensional droplet. Our experiments use ultralow contact angle hysteresis liquidlike surfaces which appear almost identical from their static wettability, but have speeds of motion which differ by more than 1 order of magnitude for the same driving force. From the force-velocity relationships, we have constructed a molecular-kinetic theory type model which incorporates the droplet-on-solid friction and through the coefficients of kinetic friction also includes the droplet adhesive force. Finally, the concepts presented are not limited to droplets, but are relevant to the motion of any three-phase contact line and to three-phase systems, such as flow in narrow capillaries and the motion of bubbles on surfaces.

Acknowledgments—We thank the UK EPSRC for financial support under Grants No. EP/V049348/1, No. EP/V049615/1, and No. EP/V049615/2. Y. W. acknowledges PhD studentship funding from the UK EPSRC.

Data availability—The data that support the findings of this Letter are not publicly available upon publication because it is not technically feasible and/or the cost of preparing, depositing, and hosting the data would be prohibitive within the terms of this research project. The data are available from the authors upon reasonable request.

- [1] T. Young, III. An essay on the cohesion of fluids, *Philos. Trans. R. Soc. London* **95**, 65 (1805).
- [2] P. G. de Gennes, Wetting: Statics, and dynamics, *Rev. Mod. Phys.* **57**, 827 (1985).
- [3] D. Bonn, J. Eggers, J. Indekeu, J. Meunier, and E. Rolley, Wetting and spreading, *Rev. Mod. Phys.* **81**, 739 (2009).
- [4] R. D. Deegan, O. Bakajin, T. F. Dupont, G. Huber, S. R. Nagel, and T. A. Witten, Capillary flow as the cause of ring stains from dried liquid drops, *Nature (London)* **389**, 827 (1997).
- [5] H. Butt *et al.*, Contact angle hysteresis, *Curr. Opin. Colloid Interface Sci.* **59**, 101574 (2022).

- [6] N. Gao, F. Geyer, D. W. Pilat, S. Wooh, D. Vollmer, H.-J. Butt, and R. Berger, How drops start sliding over solid surfaces, *Nat. Phys.* **14**, 191 (2018).
- [7] A. Laroche *et al.*, Tuning static drop friction, *Droplet* **2**, 1 (2023).
- [8] H.-J. Butt, R. Berger, J. De Coninck, and R. Tadmor, Drop friction, *Nat. Rev. Phys.* **7**, 425 (2025).
- [9] M. Backholm, D. Molpeceres, M. Vuckovac, H. Nurmi, M. J. Hokkanen, V. Jokinen, J. V. I. Timonen, and R. H. A. Ras, Water droplet friction and rolling dynamics on superhydrophobic surfaces, *Commun. Mater.* **1**, 64 (2020).
- [10] M. J. Hokkanen, M. Backholm, M. Vuckovac, Q. Zhou, and R. H. A. Ras, Force-based wetting characterization of stochastic superhydrophobic coatings at nanonewton sensitivity, *Adv. Mater.* **33**, 2105130 (2021).
- [11] X. Li, F. Bodziony, M. Yin, H. Marschall, R. Berger, and H.-J. Butt, Kinetic drop friction, *Nat. Commun.* **14**, 4571 (2023).
- [12] L. Chen, S. Huang, R. H. A. Ras, and X. Tian, Omniphobic liquidlike surfaces, *Nat. Rev. Chem.* **7**, 123 (2023).
- [13] I. J. Gresham and C. Neto, Advances and challenges in slippery covalently-attached liquid surfaces, *Adv. Colloid Interface Sci.* **315**, 102906 (2023).
- [14] I. J. Gresham, S. G. Lilley, A. R. J. Nelson, K. Koynov, and C. Neto, Nanostructure explains the behavior of slippery covalently attached liquid surfaces, *Angew. Chem., Int. Ed.* **62**, 1 (2023).
- [15] H. Barrio-Zhang, É. Ruiz-Gutiérrez, S. Armstrong, G. McHale, G. G. Wells, and R. Ledesma-Aguilar, Contact-angle hysteresis and contact-line friction on slippery liquid-like surfaces, *Langmuir* **36**, 15094 (2020).
- [16] B. Khatir, Z. Azimi Djivejin, P. Serles, T. Filleter, and K. Golovin, Molecularly capped omniphobic polydimethylsiloxane brushes with ultra-fast contact line dynamics, *Small* **2301142**, 1 (2023).
- [17] T. D. Blake and J. M. Haynes, Kinetics of liquid liquid displacement, *J. Colloid Interface Sci.* **30**, 421 (1969).
- [18] T. D. Blake, The physics of moving wetting lines, *J. Colloid Interface Sci.* **299**, 1 (2006).
- [19] G. Amontons, De la resistance causée dans les machines (About resistance and force in machines), *Mémoires de l'Académie Royale* **257**, 206 (1699), <https://gallica.bnf.fr/ark:/12148/bpt6k35013/f344.item>.
- [20] E. Popova and V. L. Popov, The research works of Coulomb and Amontons and generalized laws of friction, *Friction and wear in machinery* **3**, 183 (2015).
- [21] G. McHale, N. Gao, G. G. Wells, H. Barrio-Zhang, and R. Ledesma-Aguilar, Friction coefficients for droplets on solids: The liquid–solid Amontons’ laws, *Langmuir* **38**, 4425 (2022).
- [22] C. G. L. Furmidge, Studies at phase interfaces. 1. Sliding of liquid drops on solid surfaces and a theory for spray retention, *J. Colloid Sci.* **17**, 309 (1962).
- [23] J. J. Bikerman, The fundamentals of tackiness and adhesion, *J. Colloid Sci.* **2**, 163 (1947).
- [24] G. Macdougall and C. Ockrent, Surface energy relations in liquid/solid systems I. The adhesion of liquids to solids and a new method of determining the surface tension of liquids, *Proc. R. Soc. A* **180**, 151 (1942).

- [25] See Supplemental Material at <http://link.aps.org/supplemental/10.1103/ljbc-k193> for derivations and theoretical calculations, and materials and methods, supporting the main text, and a video showing droplet motion on SOCAL and c50-SOCAL surfaces tilted at two angles.
- [26] N. Le Grand, A. Daerr, and L. Limat, Shape and motion of drops sliding down an inclined plane, *J. Fluid Mech.* **541**, 293 (2005).
- [27] A. W. Adamson and A. P. Gast, *Physical Chemistry of Surfaces*, 6th ed. (Wiley-Blackwell New York, 1997).
- [28] L. Wang and T. J. McCarthy, Covalently attached liquids: Instant omniphobic surfaces with unprecedented repellency, *Angew. Chem., Int. Ed.* **55**, 244 (2016).
- [29] S. Armstrong, G. McHale, R. Ledesma-Aguilar, G. G. Wells, R. Ledesma-Aguilar, and G. G. Wells, Pinning-free evaporation of sessile droplets of water from solid surfaces, *Langmuir* **35**, 2989 (2019).
- [30] I. J. Gresham, H. Barrio-Zhang, J. H. Cho, B. Khatir, G. G. Wells, K. Golovin, G. McHale, and C. Neto, Comparative study on protocols to prepare slippery liquid-like PDMS coatings—pitfalls and time-savers, *Nat. Protoc.*, [10.1038/s41596-025-01253-6](https://doi.org/10.1038/s41596-025-01253-6) (2025).
- [31] C. W. Extrand and A. N. Gent, Retention of liquid drops by solid surfaces, *J. Colloid Interface Sci.* **138**, 431 (1990).
- [32] A. I. ElSherbini and A. M. Jacobi, Retention forces and contact angles for critical liquid drops on non-horizontal surfaces, *J. Colloid Interface Sci.* **299**, 841 (2006).
- [33] O. V. Voinov, Hydrodynamics of wetting, *Fluid Dyn.* **11**, 714 (1977).
- [34] R. G. Cox, The dynamics of the spreading of liquids on a solid surface. Part 2. Surfactants, *J. Fluid Mech.* **168**, 195 (1986).
- [35] T. D. Blake and J. De Coninck, The influence of solid-liquid interactions on dynamic wetting, *Adv. Colloid Interface Sci.* **96**, 21 (2002).